

AUTOKART RC

Development, Testing & Actuator Control Report

30 July 2026 - 28 September 2026

Follow-up revision covering work after 02 September 2026

Purpose

Consolidated engineering record of AutoKart firmware changes, actuator and sensor bench tests, wiring decisions, observed faults, corrective actions, and current validation status. This revision extends the 02 September baseline through 28 September.

Steering

NEMA 23/TB6600 baseline
NEMA 34 + R86mini test added
Open-loop STEP/DIR/ENA

Brake

NEMA 23 + TB6600
GPIO10/11/12
Encoder feedback under evaluation

Traction

BLD750 + BLDC
GP8630N speed command
3 V to 5 V test extension

Safety & sensing

LoRa LAY37 E-stop bench path
IMU, AS5600, AS5047P
RS485 sensor-link prototype

Scope: This document covers firmware, electronics, sensors, actuators, and physical bench/vehicle testing only.

Engineering Development Report | Updated 28 September 2026

AUTOKART RC | EXECUTIVE SUMMARY

Executive Summary

Engineering status as of 28 September 2026

After the 02 September baseline, the project added direct IMU and encoder tests, a two-controller RS485 link, LoRa emergency-stop prototypes, revised BLDC transistor pin assignments, a 5 V traction-command test, and a NEMA 34/R86mini steering bench program.

Confirmed progress

The ESP32-S3 traction bench controller runs with BRK on GPIO13, EN on GPIO14, F/R on GPIO15, and a GP8630N limit of 5.00 V. Steering is smooth with the kart jacked or at low pulse rates. The LoRa transmitter and standalone receiver tests can exchange explicit `ESTOP, 0` and `ESTOP, 1` packets.

Primary unresolved issues

Steering still stutters or slips under tyre load and begins to lose smoothness around 600-800 pulses/s in the recorded test. BLDC speed remains strongly asymmetric: counter-clockwise runs at substantially higher RPM than clockwise for the same requested voltage. The combined LoRa/BLDC implementation is not yet the default controller and still needs an end-to-end validation pass.

Key findings after 02 September

- Direct MPU-9250/6500 detection required NCS held HIGH and the tested SDA/SCL wiring moved to GPIO11/GPIO12.
- RS485 communication worked after direction control and packet framing were corrected; early logs showed fragmented and corrupted packets.
- The remote LAY37 E-stop was kept electrically separate from actuator GPIO. Its state is transmitted by LoRa from the Pico 2 side.
- BLDC EN and BRK pins were swapped in firmware to match the transistor wiring: BRK GPIO13, EN GPIO14.
- The throttle ceiling changed from 3.00 V to 5.00 V, producing a significant RPM increase.
- The web slider changed from 0.05 V to 0.01 V steps and request coalescing was added to prevent an unresponsive UI.
- AS5600 I2C, AS5047P SPI/PWM, and brake-feedback calibration utilities were added as isolated tests.
- NEMA 34 + Rtelligent R86mini testing was added on 28 September with a conservative ramp and +/-20 degree limits.

Current default versus experimental files

Purpose	File	Status
Default BLDC bench controller	bench/full_bldc_test.py	Working bench baseline; no LoRa
LoRa + BLDC E-stop integration	bench/full_bldc_lora_estop_esp32s3.py	Experimental; retest required
NEMA 34 steering	bench/nema34_test.py	Static validation passed; hardware test pending
Integrated actuator controller	firmware/ autokart_rc_v5_controlled_stop.py	Open-loop steering/brake

Engineering Development Report | Updated 28 September 2026

AUTOKART RC | DATED DEVELOPMENT TIMELINE

Development Timeline

User test log and Git history after 02 September

Date	Change / test	Evidence and result
04 Sep	Firmware repository initialized; IMU, ESP32-C3 firmware, RS485 tests and existing actuator files added.	Git commits 14fbad4 through 6e2a69f.
05 Sep	MPU-9250/6500 test corrected; AS5600 and AS5047P direct encoder tests added.	ed7b10f, 989882d, b8e478d. IMU commit recorded as working.
07 Sep 08 Sep user log	Pico 2 + RA-02 and ESP8266 + RA-02 tests added. Remote LAY37 E-stop path introduced.	Git: 58847c7, 285bd0c. User log: E-stop implemented through LoRa on 08 Sep; switch isolated from actuator controls.
21 Sep	LoRa chat, transistor pin correction, 5 V BLDC test, smooth slider patch, architecture sheet, LoRa E-stop integration and isolation tests.	Git: fec57a through 4abfe30.
21 Sep	BLDC command ceiling changed from 3 V to 5 V.	Significant RPM increase. Counter-clockwise high RPM; clockwise much lower RPM.
21 Sep	Steering disengaged from ground; pulse rate manipulated.	Smooth at low pulse rate; stuttering began around 600-800 pulses/s.
22 Sep	Steering tested jacked and unjacked.	Jacked: smooth. Unjacked/on tyre load: stuttering or slipping.
22 Sep	First follow-up firmware report generated.	Git commit faf345f.
23 Sep	AS5600 brake feedback and isolated target-angle tools added; AS5047P PWM tests added.	Git: 208b1fb, 97a04ed, 11f188e.
28 Sep	NEMA 34 + Rtelligent R86mini steering test added.	Git commit 36f4a94. Syntax and motion math validated; hardware result not yet recorded.

Date reconciliation

Git history begins on 04 September. The 08, 21 and 22 September observations above come from the project test log supplied for this revision. Git dates identify when code was committed; bench-test dates identify when hardware behavior was observed.

Current Bench Architecture

Actuator command paths

Wi-Fi browser / laptop

ESTOP,1 = stop

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SV=0, BRK applied, EN stop

The switch is electrically isolated from actuator-control GPIO: the only connection between remote and vehicle sides is the LoRa radio link. This reduces long emergency-switch wiring and prevents the handheld switch circuit from sharing actuator signal wires.

Integration limitation: The current ESP32 RA-02 bench mapping uses GPIO4/5/6 and GPIO10. Those pins overlap steering and brake controls. It is valid for the BLDC-only LoRa bench test, but it is not a conflict-free full-vehicle pinout.

Communication & Sensor Test Circuits

ESP32-C3 to ESP32-S3 RS485 bench link

XIAO ESP32-C3	CAT5e twisted pair	ESP32-S3
GPIO21 TX -> MAX3485 RX	A ===== A	MAX3485 RX <- GPIO17 TX
GPIO20 RX <- MAX3485 TX	B ===== B	MAX3485 TX -> GPIO18 RX
GPIO10 -> MAX3485 EN	GND/reference ----- GND	MAX3485 EN <- GPIO16
Direction: EN HIGH = transmit; EN LOW = receive		
UART: 9600 baud, 8-N-1		
Test protocol: PING,sequence <-> PONG,sequence		

Why these pins: XIAO GPIO20/21 correspond to exposed RX/TX labels; GPIO10 controls transceiver direction. ESP32-S3 GPIO17/18 form a separate UART pair and GPIO16 controls direction. Module labels describe module signal direction, so MCU TX connects to module RX/RXO.

Standalone encoder connections

Encoder/ interface	ESP32-S3 pins	Reason and limitation
AS5600 I2C	SDA GPIO8, SCL GPIO9	Can share I2C with GP8630N because addresses differ: 0x36

		and 0x58.
AS5047P SPI	SCK13, MOSI14, MISO15, CS16	Contiguous test map. Conflicts with BLDC GPIO13/14/15; isolated only.
AS5047P PWM	W/PWM or I/PWM -> GPIO4	Simple pulse capture after PWM enable. Conflicts with steering STEP.
MPU-9250/6500	NCS10, SDA11, SCL12	NCS HIGH selects I2C. Overlaps brake pins; standalone only.

Signal requirements

- Use a diametrically magnetized magnet centered over the encoder IC.
- AS5600 raw 0-4095 maps to 0-360 degrees.
- AS5047P SPI provides 14-bit angle; PWM must be enabled through SETTINGS1.
- All ESP32-S3 inputs must remain within 3.3 V logic limits.

ESP32-S3 Pinout Matrix

Current integrated functions and isolated-test overlaps

GPIO	Current integrated role	Bench-test role(s)	Selection rationale / conflict
4	Steering STEP	RA-02 SCK; AS5047P PWM; NEMA34 STEP	Reused for steering replacement test. Radio/encoder isolated only.
5	Steering DIR	RA-02 MOSI; NEMA34 DIR	Grouped with GPIO4/6 for STEP/DIR/ENA.
6	Steering ENA	RA-02 MISO; NEMA34 ENA	Radio conflict in full integration.
7	Unassigned	RA-02 NSS/CS	Selected to keep BLDC transistor pins free.
8	GP8630N SDA	AS5600 SDA	Shared I2C possible; unique addresses.
9	GP8630N SCL	AS5600 SCL	Shared I2C clock.
10	Brake STEP	RA-02 RESET; IMU NCS	Bench-only conflict if brake is present.
11	Brake DIR	IMU SDA	Standalone IMU overlap.
12	Brake ENA	IMU SCL	Standalone IMU overlap.
13	BLD750 BRK	AS5047P SCK	Physical transistor wiring; encoder SPI conflict.
14	BLD750 EN	AS5047P MOSI	Changed from earlier EN13/BRK14.
15	BLD750 F/R	AS5047P MISO	Traction direction; encoder conflict.

16	Available	RS485 EN; AS5047P CS	Bench direction/CS output.
17	Available	RS485 TX	Dedicated UART bench link.
18	Available	RS485 RX; RA-02 DIOo	Conflict if both links are connected.

Pin-budget conclusion: Tested pinouts are individually valid but are not yet one conflict-free vehicle pinout. A frozen GPIO allocation and power/current budget are required before full integration.

BLD750 mapping change

Signal	Before 21 Sep	Current	Reason
EN	GPIO13	GPIO14	Corrected to match transistor wiring.
BRK	GPIO14	GPIO13	Swapped with EN.
F/R	GPIO15	GPIO15	Unchanged.

Engineering Development Report | Updated 28 September 2026

AUTOKART RC | LORA E-STOP PINOUTS

LoRa E-Stop Pinouts & Logic

Pico 2 remote transmitter

Signal	Pico 2	Role
MISO	GP16	SPI receive
NSS/CS	GP17	SPI chip select
SCK	GP18	SPI clock
MOSI	GP19	SPI transmit
RESET	GP20	Radio reset
DIOo	GP21	Packet status
LAY37 NC	GP22	Switch input with 4.7 kOhm pull-up

```
Pico GP22 ---- LAY37 NC contact ---- GND

Pico GP22 ---- 4.7 kOhm resistor ---- 3.3 V

NC closed / released : GPIO22 LOW  -> ESTOP,0

NC open / pressed   : GPIO22 HIGH -> ESTOP,1
```

Broken switch wire : GPIO22 HIGH -> STOP indication

ESP32-S3 receiver for BLDC-only bench test

RA-02	ESP32-S3	Rationale
SCK	GPIO4	Avoid BLDC GPIO13/14/15.
MOSI	GPIO5	Alternate SPI.
MISO	GPIO6	Alternate SPI.
NSS/CS	GPIO7	Free chip select.
RESET	GPIO10	Free in BLDC-only test.
DIO0	GPIO18	Packet status.

Message logic change

```
state = 1 if estop.value() else 0

packet = "ESTOP,{},{}".format(state, sequence)

ESTOP,0,n -> released / clear red UI

ESTOP,1,n -> command stop and brake
```

The first design used a continuous healthy heartbeat and stopped whenever packets were delayed. Bench delays caused false stops. The bench logic changed to explicit stop/clear messages so plugging or unplugging the Pico does not independently raise the red screen. This is diagnostic behavior, not fail-safe radio-loss behavior.

Engineering Development Report | Updated 28 September 2026

AUTOKART RC | BLDC DEVELOPMENT

BLDC Control Changes

21 September: 3 V to 5 V test

```
BRK_PIN = 13

EN_PIN = 14

FR_PIN = 15
```



```
MAX_VOLTAGE = 5.00

RAMP_STEP = 0.05

RAMP_DELAY_MS = 40

COMMAND_TIMEOUT_MS = 1500
```

Change	Observation	Interpretation
SV ceiling increased from 3.00 V to 5.00 V.	Significant RPM increase.	GP8630N/BLD750 path responds to larger voltage.
Direction changed through F/R.	CCW high RPM; CW much lower RPM.	Open: check actual SV, F/R level, brake release, Hall/phase wiring, controller parameters and load.
Slider changed to 0.01 V steps with request coalescing.	More responsive while dragging.	Only latest value is sent after prior HTTP request.

Slider request logic

```
if (enRunning && brakeReleased) {

    speedRequestPending = true;

    sendLatestSpeed();

}

if (speedRequestInFlight || !speedRequestPending) return;
```

Safe-stop sequence

```
target_voltage = 0.0

write_voltage(0.0)

brake_apply()

enable_stop()
```

Validation boundary: A 5 V software limit does not prove maximum motor speed or both directions are safe. Directional RPM asymmetry remains open.

Steering Test Results

21-22 September observations

Condition	Observation	Engineering meaning
Disengaged / low pulse rate	Smooth.	Basic STEP/DIR and driver/motor wiring work at low demand.
Approximately 600-800 pulses/s	Stuttering begins.	Possible torque drop, acceleration, current limit, supply sag, resonance or binding.
Kart jacked	Smooth steering.	Mechanism moves without tyre scrub load.
Kart unjacked / tyres loaded	Stuttering or slipping.	Torque margin is below stationary tyre/load requirement or friction is excessive.

28 September NEMA 34 / R86mini configuration

```
PULSES_PER_MOTOR_REV = 3200

MOTOR_GEAR_TEETH = 17

STEERING_GEAR_TEETH = 34

GEAR_RATIO = 34 / 17          # 2.0

PULSES_PER_STEERING_DEG = 17.7778

LEFT_LIMIT_DEG  = -20.0

RIGHT_LIMIT_DEG =  20.0

START_PERIOD_US = 3000        # 333 pulses/s

FAST_PERIOD_US  = 1200        # 833 pulses/s

STEP_HIGH_US    = 10
```

Output	ESP32-S3	R86mini	Note
STEP	GPIO4	PUL	10 us active pulse.
DIR	GPIO5	DIR	30 us setup.
ENABLE	GPIO6	ENA	Code assumes optocoupler ON disables.

Electrical warning: Confirm the exact R86mini input-level specification and polarity for the installed revision. Set SW1-SW3 from the NEMA 34 nameplate current, verify coil pairs, and use the separate motor supply. Position remains open-loop.

IMU & Encoder Development

MPU-9250 / MPU-6500

Date	Problem / change	Result
04 Sep	I2C scan returned no devices on GPIO8/9. NCS/address mode uncertain.	MPU not detected.
05 Sep	SDA11, SCL12, NCS10 HIGH, ADo GND; WHO_AM_I accepts 0x71/0x70.	Committed as working IMU test.

AS5600 isolated angle processing

```
def shortest_raw_delta(current_raw, target_raw):  
  
    return ((target_raw - current_raw + 2048) % 4096) - 2048  
  
  
angle_deg = raw * 360.0 / 4096.0
```

The isolated tool prints raw data, filtered angle, magnet status, and the shortest clockwise/counter-clockwise move to an entered angle. The brake-feedback utility captures released zero and applied span, then reports 0-100% feedback. Neither tool moves a motor.

AS5047P PWM bring-up

```
SETTINGS1 = 0x0018  
  
SETTINGS1_W_PWM_ON = 0x0081  
  
  
SPI: SCK13, MOSI14, MISO15, CS16  
  
PWM: W/PWM -> GPIO4
```

W/PWM produced a high pulse followed by no low edge (high=12889, low=-2); I/PWM produced no valid edge (high=-1, low=-2). PWM was not selected/enabled. The helper now writes volatile SETTINGS1 and measures W/PWM.

Integration status: Encoder feedback is not yet merged into steering or brake loops. Integrated position remains open-loop pulse count.

Errors Faced During Bring-up

Date	Error / symptom	Cause / diagnosis	Action / status
04-05 Sep	IMU scan empty; MPU not found.	Pins/mode did not match board; NCS needed for I2C.	Moved pins, held NCS HIGH, checked ADo/WHO_AM_I.
04 Sep	RS485 split, blank or 0x00/0xFF packets.	Half-duplex direction timing and buffering.	Direction control and explicit PING/PONG parsing; CRC still needed.
04-08 Sep	mpremote ... may be in use.	VS Code/Thonny/Electron or Python owned serial.	One serial client; inspect with lsof/fuser.
07 Sep	Pico RP2 Boot but no ttyACM.	UF2 bootloader mass-storage mode.	Copy correct Pico 2 UF2, reboot, rediscover port.
07 Sep	Thonny Python: no module mpremote.	Different Python environment.	Activate project Python; avoid bundled Thonny interpreter.
07-08 Sep	ESP8266 raw REPL failure; ttyUSBo locked.	Running main.py or another process owned port.	Stop loop, release port, reset and reconnect.
21 Sep	ImportError: no module named network.	ESP32 file executed on Pico/wrong ACM device.	Identify board by platform, not ACM number.
21 Sep	LoRa heartbeat delay repeatedly stopped BLDC.	Timeout too sensitive to packet latency/loss.	Bench sender changed to explicit ESTOP,0/1; link-loss policy open.
21 Sep	name 'reason' isn't defined.	Safe-stop code displaced during merge.	Restored default non-LoRa BLDC; kept LoRa separate.
21 Sep	Pico TX visible; ESP32 did not process packets.	Radio processing and web/control loop not reliably integrated.	Standalone one-shot TX/RX tests added.
21 Sep	BLDC slider unresponsive.	Overlapping HTTP requests while dragging.	One request in flight; latest value pending; 0.01 V step.
21 Sep	CW/CCW RPM differed greatly.	F/R, controller, wiring, load or actual SV difference.	Open; measure both directions.

Errors Faced During Bring-up

Continued

Date	Error / symptom	Cause / diagnosis	Action / status
21-22	Steering smooth jacked;	Torque margin, current, voltage sag,	Open. NEMA34/R86mini

Sep	stutters/slips unjacked and near 600-800 pulses/s.	acceleration, resonance or friction.	path added.
23 Sep	AS5047P W/PWM high/no low edge; I/PWM no edges.	PWM not selected/enabled; output remained UVW/ABI.	SPI helper writes volatile SETTINGS1=0x0081.
28 Sep	Could not overwrite main.py; old AS5047 prompt blocked raw REPL.	Old input loop caught Ctrl-C.	Enter q, return to >>>, exit Ctrl-], then copy NEMA file.
Ongoing	ttyACM0/ttyACM1 identities changed.	Linux assigns names dynamically.	Rediscover ports; do not infer board from ACM number.

Safety trade-off

Explicit-message E-stop mode prevents false red-screen activation when the remote is unplugged, as requested for bench testing. It also means radio loss alone does not command a stop. A vehicle design must distinguish STOP, CLEAR, stale sequence, corrupted packets and lost communication, with an independent hardware stop path.

Why the stable BLDC file was restored

Commit 4abfe30 removed the partial LoRa merge from `full_bldc_test.py`. This preserved known-working traction behavior while retaining LoRa work in separate files.

Diagnostic principle: isolate one interface, verify identity or signal edges, then integrate. Serial access proves MCU communication only; it does not prove the radio, sensor, actuator or mechanical load.

Current Validation Matrix

Subsystem	Evidence	Status	Next gate
BLDC bench	Stable after pin correction; 5 V increased RPM.	Bench working	Resolve direction asymmetry; measure SV/RPM.
NEMA23/TB6600 steering	Smooth jacked/low rate; stutters under load.	Loaded test failed	Measure torque, current, voltage and friction.
NEMA34/R86mini	Code/math validated; no powered result recorded.	Pending	Confirm interface/current/coils; unloaded 1 degree.
Brake axis	Open-loop STEP/DIR/ENA logic.	Pending load test	Add feedback and end stops.
LoRa E-stop	Standalone packets and button transmission observed.	Partial	Integrated STOP/CLEAR/link-loss test.
RS485	PING/PONG observed; early corruption.	Bench link	Framing, CRC, sequence, termination.
MPU-9250/6500	Working I2C test committed.	Standalone working	Bias/orientation calibration and final pins.
AS5600	I2C and isolated angle utilities.	Utility ready	Capture real zero/span under load.
AS5047P	SPI test and PWM-enable helper; timeout diagnosed.	Recheck PWM	Verify valid frame and magnet geometry.

Recommended order of work

- Freeze a conflict-free ESP32-S3 pin map.
- Complete unloaded NEMA34 tests at 1 degree, 5 degrees and center.
- Measure loaded steering torque/current/voltage.
- Measure BLDC SV and RPM in CW and CCW.
- Validate encoder feedback, then close steering/brake loops.
- Reintegrate LoRa into a copy of stable BLDC and test STOP, CLEAR, stale packets and link loss.
- Add limit switches and an independent hardware E-stop before vehicle operation.

Source Files & Git Traceability

Principal source files

Area	Source
Traction bench	<code>bench/full_bldc_test.py</code>
LoRa traction/E-stop	<code>bench/full_bldc_lora_estop_esp32s3.py</code>
Pico transmitter	<code>bench/pico2_ra02_estop_transmitter.py</code>
NEMA34 steering	<code>bench/nema34_test.py</code>
Integrated actuators	<code>firmware/autokart_rc_v5_controlled_stop.py</code>
IMU	<code>bench/imu_esp32s3_test.py</code>
AS5600	<code>bench/as5600_esp32s3_test.py</code> ; <code>as5600_isolated_test.py</code> ; <code>as5600_brake_feedback_test.py</code>
AS5047P	<code>bench/as5047p_esp32s3_test.py</code> ; <code>as5047p_pwm_isolated_test.py</code> ; <code>as5047p_enable_w_pwm_test.py</code>
RS485	<code>bench/rs485_c3_test.py</code> ; <code>rs485_s3_test.py</code> ; <code>docs/RS485_CONNECTIONS.md</code>

Key Git commits after 02 September

Commit	Date	Change
<code>ed7b10f</code>	05 Sep	Working IMU
<code>989882d</code>	05 Sep	Working encoders
<code>58847c7</code>	07 Sep	Pico 2 - RA02
<code>feca57a</code>	21 Sep	LoRa test
<code>bb748ab</code>	21 Sep	Transistor pin correction
<code>41a757d</code>	21 Sep	5 V and slider patch
<code>9b48efb</code>	21 Sep	LoRa E-stop
<code>4abfe30</code>	21 Sep	Default BLDC without LoRa
<code>208b1fb</code>	23 Sep	AS5600 feedback
<code>11f188e</code>	23 Sep	AS5047P PWM
<code>36f4a94</code>	28 Sep	NEMA34 test

Report basis: repository history through commit `36f4a94`, plus dated bench observations supplied for o8,

21 and 22 September. Hardware observations are separated from static code validation.